

ANALYSTLAB AFRICA

EXPERIENCE LAB INTERNSHIP PROGRAMME

WEEK 1 PROJECT

FinTrust Digital Bank

Track: Data Analytics

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Week: 1

Project: FinTrust Financial Intelligence & Digital Banking Support
Solution

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PART 1: BUSINESS UNDERSTANDING

1.1 Introduction

FinTrust Digital Bank is a fictional digital banking Organization seeking to use data, machine learning and generative artificial intelligence to improve its understanding of customer and transaction behaviour, support risk-related decision-making, improve operational processes and strengthen digital customer support.

As a Data Analytics intern, my role is to determine how FinTrust's customer and transaction data can be transformed into useful business intelligence that supports better decision-making.

Week 1 focuses on understanding the business problem, reviewing the available resources, identifying important business questions, defining relevant KPIs and planning the analysis and dashboard development for the following weeks.

1.2 What Business Problem Is FinTrust Trying to Solve?

FinTrust needs to better understand how its customers interact with the bank and how transactions are performed across different transaction types, channels and customer segments.

The organization also needs better visibility into transaction performance and risk-related patterns. Without structured analysis, management may find it difficult to identify changes in customer behavior, transaction activity, transaction values, failed or reversed transactions, and transactions requiring risk review.

The project therefore aims to transform available customer and transaction data into useful insights that can support business intelligence and informed decision-making.

1.3 Why Is the Problem Important?

Understanding customer and transaction behavior is important because digital banks generate large amounts of data through customer accounts and transactions.

Effective analysis can help FinTrust:

- Understand its customer base and customer segments.
- Monitor transaction activity and transaction values.

- Identify the most frequently used transaction types and channels.
- Monitor successful, failed, reversed and pending transactions.
- Identify patterns in transactions requiring risk review.
- Support evidence-based business decisions.
- Provide management with a clearer view of the bank's digital activity.

1.4 How Can the Data Analytics Track Contribute?

The Data Analytics track can contribute by:

- Profiling and understanding the available customer and transaction datasets.
- Identifying meaningful business and analytical questions.
- Calculating relevant KPIs.
- Analyzing customer and transaction patterns.
- Developing an interactive Power BI dashboard.
- Communicating findings and business insights clearly to decision-makers.

1.5 Expected Output

The expected Data Analytics outputs include:

- Business understanding documentation.
- Data profiling and understanding report.
- Analytical questions.
- KPI definition table.
- Dashboard wireframe.
- Initial analysis plan.
- A future Power BI dashboard containing relevant KPIs, visuals, filters and business insights.

PART 2: RESOURCE REVIEW

2.1 Introduction

The approved FinTrust resources provided for the project include three data resources, a GenAI resource and project resources.

The resources relevant to the Data Analytics track are reviewed below.

2.2 Data Resources

1. FinTrust Customer Data

The FinTrust Customer Data contains customer-level information such as customer identification, demographic information, account information, customer segment, tenure, digital engagement and preferred banking channel.

How it will be used:

The dataset will be used to understand the characteristics of FinTrust's customer base and to analyze differences in behaviour across customer segments, account types and other customer characteristics.

Limitations:

The data is synthetic and created for educational purposes. It therefore should not be interpreted as real customer information or used to make claims about actual banking customers.

2. FinTrust Transaction Data

The FinTrust Transaction Data contains transaction-level information, including transaction ID, customer ID, transaction date/time, transaction type, amount, channel, device type, location, international transaction status, transaction status and risk-review information.

How it will be used:

It will be used to analyze transaction activity, transaction value, transaction channels, transaction status and risk-related patterns.

Limitations:

The transaction data is synthetic and contains a small number of missing values in some fields. These limitations will need to be considered during later data preparation and analysis.

3. FinTrust Data Dictionary

The FinTrust Data Dictionary provides definitions and descriptions of the fields contained in the FinTrust datasets.

How it will be used:

The Data Dictionary will be used alongside the actual datasets to understand the meaning and intended use of individual fields, confirm field classifications and support correct interpretation of variables during analysis.

For example, it helps provide context for fields such as Customer-ID, Digital-Engagement-Score, Risk-Review-Flag and other customer and transaction variables.

Limitations:

The Data Dictionary provides definitions for the supplied synthetic project data. It does not replace the need to inspect the actual dataset values and perform data-quality checks.

2.3 Other Approved Resources

FinTrust Financial Knowledge Base

This is a GenAI resource intended primarily for the Generative AI track. It contains approved FinTrust financial information that can support the future development of a grounded customer-support assistant.

For the Data Analytics track, it is not a primary analytical dataset, but awareness of its existence is useful when considering the wider FinTrust solution.

FinTrust Project Brief

The Project Brief provides information about the overall FinTrust project, its business context and expected solution.

It will be used to maintain alignment between the Data Analytics work and the overall project objectives.

FinTrust 4-Week Master Roadmap

The Master Roadmap provides the overall project structure and helps establish what is expected across the four-week project.

It will be used to plan the Data Analytics activities for Weeks 2–4 and understand how the Week 1 foundation connects to later project work.

PART 3: DATA UNDERSTANDING / DATA PROFILING

For this part, I reviewed the two FinTrust datasets — Customer Data and Transaction Data. I also used the FinTrust Data Dictionary to understand what the different fields mean and how they should be interpreted.

3.1 Customer Data

The FinTrust Customer Data contains 1,500 customer records and 12 columns.

The main fields include:

- Customer_ID
- Customer_Name
- Age
- Gender
- City
- Customer Segment
- Account Type
- Tenure Months
- Digital Engagement Score
- Monthly Income Band
- Preferred Channel
- Account Status

The numerical fields include Age, Tenure Months and Digital Engagement Score.

The categorical fields include Gender, City, Customer Segment, Account Type, Monthly Income Band, Preferred Channel and Account Status.

There are no missing values in the Customer Data.

3.2 Transaction Data

The FinTrust Transaction Data contains 12,000 transaction records and 11 columns.

The fields include:

- Transaction ID
- Customer ID
- Transaction Date -Time
- Transaction Type
- Amount NGN
- Channel
- Device Type
- Location
- International Transaction
- Transaction Status
- Risk Review Flag

The main numerical field is Amount NGN, while Transaction Date-Time is the date/time field.

The categorical fields include Transaction Type, Channel, Device Type, Location, International Transaction, Transaction Status and Risk Review Flag.

The transaction data has 96 missing values in Device Type and 96 missing values in Location. These are the main missing-value issues identified at this stage.

3.3 Relationship Between the Datasets

The two datasets are connected through the Customer ID field.

The Customer Data contains customer information, while the Transaction Data contains information about the transactions made by those customers.

The relationship is one-to-many, because one customer can have multiple transactions.

I also observed that all 12,000 transactions have matching Customer IDs, and all 1,500 customers appear in the transaction data.

3.4 Data Quality Observations

During the initial review, I did not find duplicate Customer IDs or Transaction IDs.

The transaction amounts also contain no negative or zero values.

The customer ages range from 18 to 65, while customer tenure ranges from 1 to 96 months.

The transaction dates cover the period from January 1, 2026 to March 31, 2026.

The main data-quality issues identified are the missing values in Device_Type and Location.

At this stage, I am only identifying and documenting the data issues. Extensive data cleaning will be carried out in the later stages of the project, as required by the Week 1 assignment.

3.3 Data Dictionary Review

I also reviewed the **FinTrust Data Dictionary** alongside the two datasets. The Data Dictionary helped me understand the meaning of the fields, the type of information each field contains, and how the fields should be interpreted during the analysis.

For example, it helped me understand fields such as **Digital Engagement Score, Customer Segment, Transaction Status and Risk Review Flag** before using them for analysis.

The Data Dictionary will also serve as a reference during the data cleaning, analysis and dashboard stages. This will help reduce the risk of misunderstanding the meaning of any field or using a field incorrectly.

Since Week 1 is mainly about understanding and planning, I have used the Data Dictionary to guide my initial review rather than making major changes to the data at this stage.

PART 4: ANALYTICAL QUESTIONS

Based on the business objectives and the available data, the following analytical questions will guide the Data Analytics work:

1. Transaction Activity

What is the total number and total value of transactions, and how do they change over time?

2. Transaction Types

Which transaction types have the highest transaction volume and transaction value?

3. Transaction Channels

Which transaction channels are most frequently used by customers?

4. Customer Behaviour

How does transaction activity differ across customer segments and account types?

5. Transaction Status

What proportion of transactions are successful, failed, reversed or pending?

6. Risk-Related Patterns

What percentage of transactions are flagged for risk review, and what patterns can be identified among flagged transactions?

7. International Transactions

How do transaction amounts differ between international and non-international transactions?

8. Digital Engagement

Is there a relationship between customer digital engagement and their transaction activity or preferred channel?

These questions cover customer behaviour, transaction activity, transaction value, transaction channels, transaction status and risk-related patterns.

PART 5: KPI PLANNING

For this project, I selected five key performance indicators (KPIs) that will help me understand FinTrust's customer base, transaction activities, transaction value, transaction performance, and risk-related activities.

KPI	Definition	Why It Matters	Required Data
Total Customers	The total number of unique customers in the dataset.	This will help me understand the size of FinTrust's customer base.	Customer ID
Total Transactions	The total number of transactions recorded in the dataset.	This will show the overall level of transaction activity among FinTrust customers.	Transaction ID
Total Transaction Value	The total amount of money involved in all recorded transactions.	This will help me understand the overall monetary value of the transactions carried out.	Amount NGN
Transaction Success Rate	The percentage of transactions that were successfully completed.	This will help me assess how well transactions are being completed and identify unsuccessful transactions.	Transaction ID, Transaction Status
Risk Review Rate	The percentage of transactions that were flagged for risk review.	This will help me understand how often transactions are being flagged and identify areas that may need further investigation.	Transaction ID, Risk Review Flag

KPI Calculation Approach

During the analysis stage, I will calculate the KPIs using the relevant fields in the customer and transaction datasets.

- **Total Customers** = Count of unique Customer IDs
- **Total Transactions** = Count of Transaction IDs
- **Total Transaction Value** = Sum of transaction amounts
- **Transaction Success Rate** = $\text{Successful Transactions} \div \text{Total Transactions} \times 100$
- **Risk Review Rate** = $\text{Risk-Flagged Transactions} \div \text{Total Transactions} \times 100$

These KPIs will later be used in the Power BI dashboard to give management a quick view of the overall performance of the FinTrust digital banking platform. They will also support the analytical questions and help identify patterns that may be useful for decision-making.

The calculations will be validated during the data analysis and dashboard development stages to ensure that the KPIs are accurate.

PART 6: DASHBOARD PLANNING / WIREFRAME

6.1 Dashboard Objective

The proposed Power BI dashboard will provide management with a simple and interactive overview of FinTrust's customer and transaction activity.

The dashboard will combine KPI cards, charts and filters to help users identify important trends and patterns.

6.2 Proposed Dashboard Layout

Top Section — KPI Cards

The dashboard will contain five KPI cards:

1. **Total Customers**
2. **Total Transactions**
3. **Total Transaction Value**
4. **Transaction Success Rate**
5. **Risk Review Rate**

Middle Section — Transaction Activity

Visual 1: Transactions by Month

Purpose: To show how transaction activity changes over time.

Visual 2: Transactions by Transaction Type

Purpose: To identify the transaction types with the highest activity.

Lower Section — Channel and Status

Visual 3: Transactions by Channel

Purpose: To understand which channels customers use most frequently.

Visual 4: Transaction Status

Purpose: To compare Successful, Failed, Reversed and Pending transactions.

Visual 5: Risk Review Rate by Customer Segment

Purpose: To compare risk-review activity across customer segments.

6.3 Proposed Filters

The dashboard will include filters for:

- Transaction Date
- Customer Segment
- Transaction Type

These filters will allow users to interact with the dashboard and explore specific areas of the data.

6.4 Expected Insights

The dashboard is expected to help management:

- Understand overall transaction activity.
- Identify changes in transaction activity over time.
- Determine the most frequently used transaction types.
- Understand customer channel preferences.
- Monitor transaction performance.
- Compare customer segments.
- Identify patterns in transactions requiring risk review.

The exact visuals and filters may be refined during later analysis if the data reveals more useful ways of presenting the findings.

PART 7: INITIAL ANALYSIS PLAN — WEEKS 2 TO 4

Week 2 — Data Preparation and Exploration

Planned Activities

- Review the customer and transaction datasets again.
- Validate data types.
- Check missing values.
- Check duplicates.
- Validate the relationship between the datasets.
- Investigate the missing Device Type and Location values.
- Perform appropriate data cleaning and preparation.
- Explore basic customer and transaction patterns.
- Use Excel, SQL or Power BI where appropriate.

Expected Output

A clean and analysis-ready dataset together with initial exploratory findings.

Week 3 — Data Analysis and KPI Calculation

Planned Activities

- Analyze transaction activity over time.
- Analyze transaction types.
- Analyze transaction channels.
- Analyze transaction status.
- Compare customer segments and account types.
- Analyze transaction values.
- Examine international versus non-international transactions.
- Analyze risk-review patterns.
- Calculate and validate the proposed KPIs.

- Identify important business insights.

Expected Output

Validated KPI calculations, analytical findings and a set of business insights to communicate through the dashboard.

Week 4 — Dashboard Development and Insights

Planned Activities

- Develop the Power BI dashboard based on the Week 1 wireframe.
- Create the five KPI cards.
- Build the planned charts and visuals.
- Add appropriate filters.
- Test interactions and calculations.
- Identify the most important insights.
- Improve dashboard clarity and presentation.
- Prepare the final analytical output.

Expected Output

An interactive Power BI dashboard containing relevant KPIs, visuals, filters and business insights.

PART 8: TRACK OBJECTIVES

The main objectives of the Data Analytics track are:

1. **Understand and profile** the FinTrust customer and transaction datasets.
2. **Identify meaningful business questions** that can be answered using the available data.
3. **Define and calculate relevant KPIs** for monitoring customer and transaction performance.
4. **Analyze customer behaviour, transaction activity and risk-related patterns** to identify useful business insights.
5. **Develop a clear Power BI dashboard** that communicates the findings to FinTrust decision-makers.

PART 9: SUCCESS CRITERIA

The Data Analytics work will be considered successful if:

1. The customer and transaction datasets are correctly understood and prepared for analysis.
2. The relationship between the datasets is correctly established and used.
3. The analysis provides answers to the identified business and analytical questions.
4. The proposed KPIs are correctly defined and calculated.
5. The Power BI dashboard presents information clearly and accurately.
6. Users can interact with the dashboard using appropriate filters.
7. The analysis identifies useful customer, transaction and risk-related patterns.
8. The resulting insights can support evidence-based decision-making.

PART 10: RISKS AND DEPENDENCIES

10.1 Potential Risks

Risk	Potential Impact	Proposed Response
Missing Device Type values	May affect channel/device analysis	Investigate and handle during Week 2 data preparation
Missing Location values	May affect location-based analysis	Assess the missing records before analysis
Incorrect data types	Could affect calculations and visualization	Validate data types during preparation
Incorrect interpretation of fields	Could lead to misleading analysis	Refer to the Data Dictionary
Incorrect KPI calculations	Could produce inaccurate management information	Validate calculations before dashboard development
Technical issues with analytical tools	Could delay analysis	Use appropriate alternative tools where necessary
Misinterpretation of synthetic risk labels	Could lead to inappropriate conclusions	Clearly treat Risk Review Flag as synthetic educational data

10.2 Future Dependencies

The Data Analytics track may depend on outputs from other tracks as the project progresses.

However, there is **no formal cross-track collaboration requirement in Week 1**. Formal collaboration begins from Week 2.